

Agentic AI Framework for Risk Scoring, Model Governance, and Monitoring

A LangGraph Multi-Agent Workflow for Data Validation, Model Scoring, Performance Monitoring, Human-in-the-Loop Review, and Executive Decision Support

Jingru Chen | 2026-07-07

Executive Summary



THE PROBLEM

Risk scoring and model governance involve multiple disconnected processes, making it difficult to ensure consistent monitoring, timely issue resolution, and transparent decision support.



THE SOLUTION

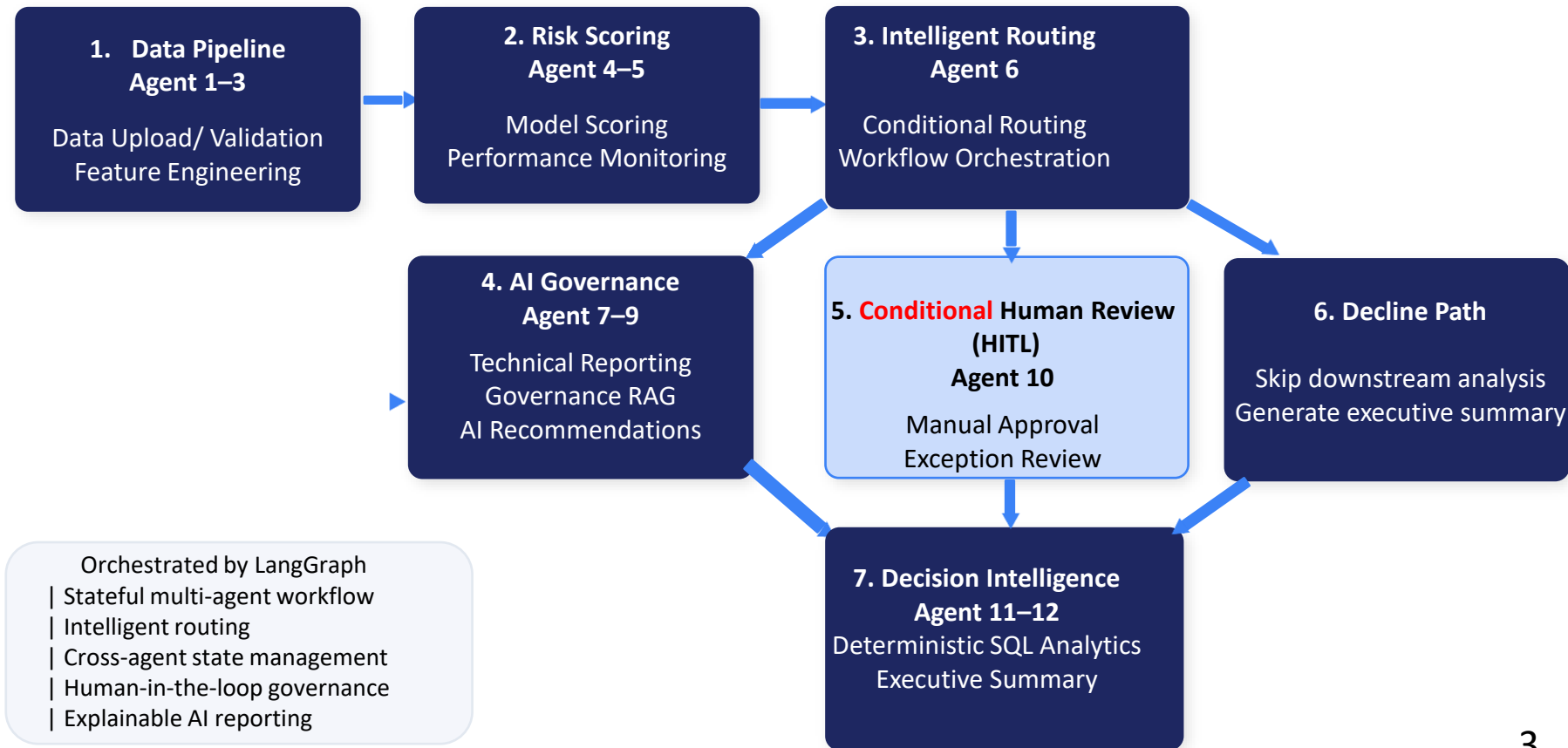
A LangGraph-based Agentic AI framework that orchestrates AI and deterministic agents to automate end-to-end risk scoring, model governance, monitoring, and executive decision support.



THE IMPACT

- 1) Automates end-to-end model governance workflows.
- 2) Reduces manual review and operational effort.
- 3) Improves transparency, explainability, and auditability.
- 4) Supports AI-assisted recommendations with human-in-the-loop review.
- 5) Generates executive-ready technical and business summaries.

LangGraph Agent Architecture



End-to-End Risk Scoring Workflow

1

Data Preparation

Data Upload, Validation, and Feature Engineering

Prepares high-quality, model-ready datasets through automated data ingestion, validation, and feature engineering.

2

Risk Scoring

Executes machine learning models to generate probability-of-default (PD) scores and prediction outputs.

3

Performance Monitoring

Evaluates AUC, KS, Precision, Recall, F1, calibration, and other monitoring metrics to assess model quality.

4

Intelligent Routing

Routes workflows dynamically based on validation, scoring, and monitoring outcomes.

5

Decision Intelligence

Generates governance recommendations, deterministic SQL analytics, and executive-ready summaries.

LangGraph Capabilities Used

Multi-Agent Orchestration

Stateful orchestration of specialized AI and deterministic agents across the complete risk scoring lifecycle.

Intelligent Workflow Routing

Conditional routing automatically directs workflows to AI governance, human review, or executive reporting based on model evaluation results.

Explainable Decision Support

Combines governance RAG, deterministic SQL analytics, and executive summaries to provide transparent, auditable decision support.

Roadmap & Next Steps

Phase 1 (POC — Now)

- ✓ LangGraph multi-agent workflow
- ✓ Automated data validation
- ✓ Feature engineering pipeline
- ✓ ML risk scoring
- ✓ Performance monitoring
- ✓ Conditional workflow routing
- ✓ Governance RAG
- ✓ Human-in-the-loop review
- ✓ Deterministic SQL analytics
- ✓ Executive reporting

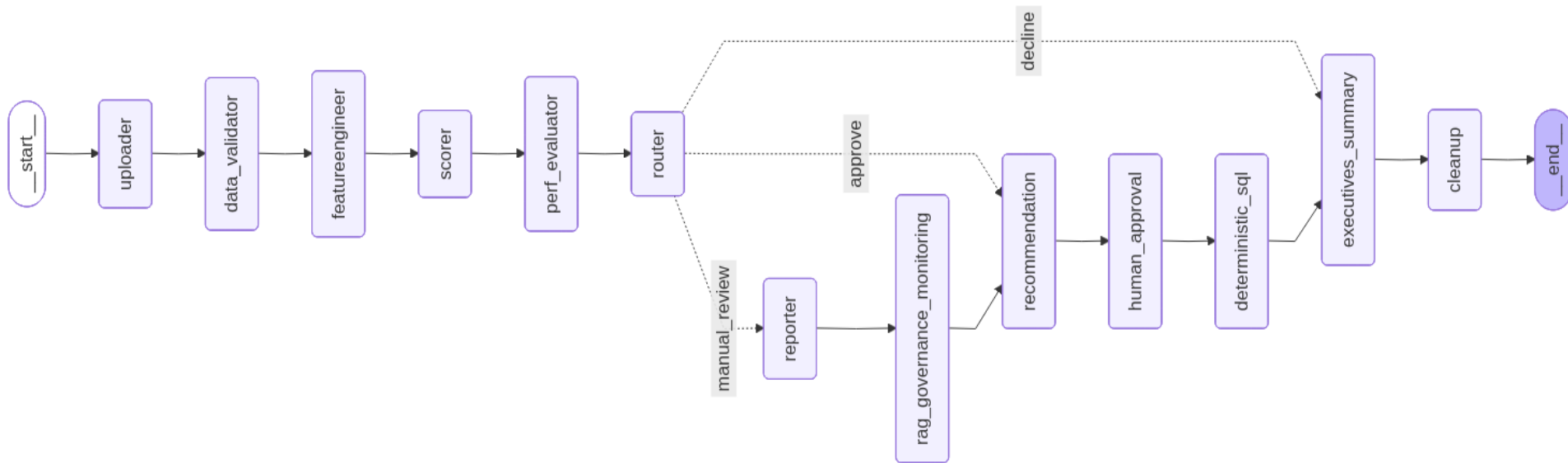
Phase 2 Production AI Governance Platform

1. Enterprise knowledge base (policies, model documentation)
2. Vector database for governance RAG
3. Multi-source data connectors
4. Explainable AI (SHAP/LIME)
5. Counterfactual analysis
6. Champion–Challenger monitoring
7. Model drift detection
8. Automated governance alerts
9. Workflow audit trail
10. Enterprise approval workflow
11. Agent memory & long-running workflows

Phase 3 Enterprise Decision Intelligence Platform

1. Real-time risk scoring
2. AI policy engine
- 3. Enterprise model governance**
- 4. Agent guardrails**
- 5. Regulatory compliance**
- 6. MLOps / LLMops integration**
7. Continuous monitoring
8. Human approval workflows
9. Multi-model orchestration

Appendix 1: LangGraph Workflow for Risk Scoring & Model Governance



Routing logic:

Manual Review → Technical Report → Governance RAG → Recommendation

Approve → Recommendation → Human Approval

Decline → Executive Summary

Appendix 2: Sample Outputs — Data Pipeline & Risk Scoring



Agentic AI Credit Risk Decisioning Platform

Agent 1 — Data Upload

	Item	Value
0	rows	67957.00
1	columns	45.00
2	memory_mb	37.92

	Validation	Status	Details
0	File exists	PASS	/content/drive/MyDrive/Colab Notebooks/CCAR_Mo...
1	Empty file	PASS	29210174 bytes



Agent 3 — Feature Engineering

	Item	Value
0	rows	67957.00
1	columns	45.00
2	memory_mb	37.92



Agent 2 — Data Validation

	Validation	Status	Details
0	Required columns	PASS	11 required columns found
1	Duplicate Rows	PASS	No duplicates
2	Missing values	WARNING	default_date: 53.1%, default_yrmo: 53.1%
3	YRMO distribution	PASS	49 unique months
4	Target distribution	PASS	Bad rate = 2.50%
5	Outliers	WARNING	5227 values outside 1%-99% range



Agent 4 — Model Scoring

	Vintage	Status	Count	Avg_PD	Precision	Recall	F1	TP	FN	TN	FP
0	201604	Success	1179	0.332076	0.033058	0.181818	0.055944	4	18	1040	117

	Item	Value
0	rows	1179
1	columns	47
2	memory_mb	0.67
3	status	Success

Appendix 3: Sample Outputs- Performance Monitoring & Decision Routing

Agent 5 — Performance Evaluation

	Metric	Status	Value	Details
0	AUC	WARNING	0.6388	
1	KS	PASS	0.2517	
2	Precision	INFO	0.0331	
3	Recall	INFO	0.1818	
4	F1	INFO	0.0559	
5	Actual Bad Rate	INFO	0.0187	
6	Predicted Bad Rate	INFO	0.1026	
7	Average PD	INFO	0.3321	

Agent 7 — Technical Report

	Validation PASS	Validation WARNING	Validation FAIL	Scoring Status	Vintage	Record Count	Average PD	Precision	Recall	F1	Route Decision	Route Reason
0	4	2	0	Success	201604	1179	0.332076	0.033058	0.181818	0.055944	RAG_AGENT	None

Agent 8 — RAG-Goverance-Monitoring

If model recall drops below threshold, investigate data drift, feature distribution shift, and target definition changes. If AUC or KS deteriorates materially, perform model monitoring review and consider recalibration. If data validation fails, stop scoring and escalate to data owner. If predicted bad rate deviates materially from actual bad rate, review calibration and cutoff strategy.

Agent 6 — Conditional Router

	Item	Value
0	route_decision	RAG_AGENT
1	route_reason	Model performance warning
2	route_result	RAG_AGENT

Note (*): This example was routed to the **Manual Review** path because the model performance triggered a warning. As a result, **Agent 7 (Technical Report)** and **Agent 8 (Governance RAG)** were triggered.

Appendix 4: Sample Outputs- Decision Intelligence & Executive Reporting

Agent 9 — Recommendation

	Issue	Status	Value	Severity	Recommendation	Human Approval Required
0	AUC	WARNING	0.6388	MEDIUM	Continue with caution. Review score distributi...	Yes

Agent 10 — Human Approval

	Approval Status	Reason
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0	PENDING_HUMAN_APPROVAL	Human approval is required before final report...
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Agent 11 — SQL Agent

	report_yrmo	metric_value
0	201511	44488971.03
1	201512	44017922.07
2	201601	43608113.70
3	201602	42695668.11
4	201603	42343606.80
5	201604	41620371.74

Agent 12 — Executive Summary

Executive Summary

0	The scoring pipeline completed for vintage 201...
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 **Workflow Completed Successfully**

Thank You

Questions & Discussion
